



Q.1. Answer the following short questions: (6x5=30)

- Explain the use of extraneous information in linear regression.
- Discuss the methodology of Econometric research.
- Under what reasons an error term is introduced in Econometric model?
- Discuss Restricted Least Squares method.
- Write short note on Stepwise regression.
- Discuss the assumptions of MLE method for GLR model.

Answer the following questions. (3x10=30)

Q.2. Show that least squares unbiased estimate of variance of disturbance term is given by

$$\hat{\sigma}_u^2 = \frac{\sum e_i^2}{n - k - 1}$$

Where e_i the i th residual, k is the number of explanatory variables of the GLR model

- Q.3. Consider the General linear regression model $\underline{Y} = \underline{X}\underline{\beta} + \underline{\varepsilon}$ which fulfills the OLS assumptions. Develop the testing procedure for $H_0: \underline{\beta} = \underline{\beta}_0$.
- Q.4. For the Simple linear regression model, show that the maximum likelihood and OLS estimators of β are equal.

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